Corner joint

Two metal sheets kept at 90^0 to each other are welded by this joint. This method is adopted when making boxes and tanks. Corner joints are adopted for thin and thick sheets.

Flange joint

The plates of the joint may be kept parallel or at 90^0 to each other. The edges of the plates are bent to form the shape of a flange.

7.20 Safety precautions for welding

We make use of electrical devices and inflammable gases like oxygen and acetylene in welding. If proper care is not shown in handling them, there is always a possibility of accidents taking place. So, welding process should be carried out with due safety and caution.

Safety precautions for gas welding

1. Gas cylinders should be kept in ventilated locations.
2. Cylinders should not be kept near hot locations. They should be kept away from electrical terminals.
3. Pressure regulators should be closed after the welding work is completed.
4. Regulators should be handled properly.
5. Old and worn out regulators should be replaced immediately.
6. The operator should wear goggles, gloves, apron and proper footwear.
7. Fire extinguishers and first – aid box should be kept ready always.

Safety precautions for arc welding

1. The welder should always wear goggles. Gloves, apron and proper footwear during welding
2. Welding shop should be located properly so that it does not cause any discomfort to others.
3. The welder should act carefully against electric shocks.
4. A high quality electrode holder should be put into use.
5. Power supply should be provided at required voltage and uniform current
6. Fire extinguishers and First – aid box should always be kept ready in a welding shop.

ACTIVITY

1. Students to visit the welding workshop to study different types of welding joints and give practice any two types of welding joints.

Questions

Part I.

Choose the correct option 1 Mark

1. Carbon Arc welding is related to .....
   a) Gas welding
   b) Resistance welding
   c) Arc welding
   d) Solid state welding
2. The heat generated during arc welding is
   a) 100^0C – 150^0C b) 50^0C – 100^0C
   c) 5000^0C-6000^0C d) 15^0 – 200^0C

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